Optimal Control - Final Project

Missile Guidance Using LQG Framework

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Overview and Problem Setup

The goal of this project is to design an optimal guidance law for a missile pursuing a maneuvering target. We employ the LQG (Linear Quadratic Gaussian) approach, which handles both the optimization of the control law and the estimation of unmeasured states from noisy observations. The engagement is analyzed in a planar setting using the classical guidance triangle geometry.

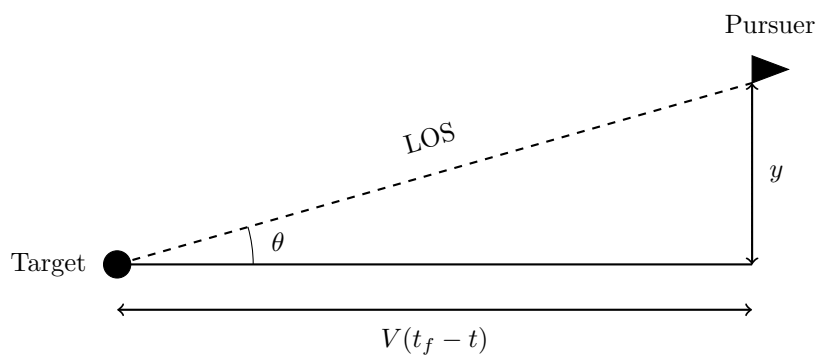

Figure 1: Planar engagement geometry between pursuer and target.

Consider a pursuer closing in on a target with constant velocity V. At any time t, the remaining range is $V(t_f - t)$, where t_f is the intercept time. The instantaneous LOS angle θ is small when the pursuer is far from the target, but grows rapidly as $t \to t_f$. Since the linearized model $\theta \simeq y/V(t_f - t)$ assumes small angles, the model becomes increasingly inaccurate near intercept; nevertheless, the given formulation is used throughout this project.

Mathematical Model

Governing Equations

The engagement kinematics are captured by three coupled stochastic differential equations:

$$dy = v\,dt, \qquad dv = (a_P - a_T)\,dt, \qquad da_T = -\frac{1}{\tau} a_T\,dt + d\beta$$

The physical interpretation of each quantity is as follows:

- y - perpendicular displacement from the initial line-of-sight
- v - rate of change of the lateral displacement

- a_P - commanded lateral acceleration of the pursuer (control variable)
- a_T - evasive acceleration of the target, modeled stochastically
- τ - time constant characterizing the target's maneuver persistence
- β - Wiener process with intensity W, representing random disturbances

Matrix Formulation
Defining the state vector $x_t = \begin{bmatrix} y & v & a_T \end{bmatrix}^T$ and control $u_t = a_P$, the system takes the standard linear stochastic form:

$$dx_t = (Fx_t + Bu_t)\,dt + G\,d\beta_t$$

with matrices:

$$F = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & -1 \\ 0 & 0 & -1/\tau \end{bmatrix}, \qquad B = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \qquad G = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

The structure reveals that: F couples the states through the kinematic chain, B shows that control directly affects only the velocity equation, and G indicates that stochastic disturbances enter solely through the target's acceleration.

Observation Equation

The pursuer measures the LOS angle θ to the target, but this measurement is corrupted by noise due to sensor imperfections and atmospheric effects. The observation process is modeled as:

$$dz_t = \theta_t\,dt + dm_t = H_t x_t\,dt + dm_t$$

The observation matrix H_t relates the state vector to the measured angle:

$$H_t = \begin{bmatrix} \frac{1}{V(t_f - t)} & 0 & 0 \end{bmatrix}$$

The term m_t represents the measurement noise, modeled as a Brownian motion with time-varying intensity $M(t)$:

$$M(t) = R_1 + \frac{R_2}{(t_f - t)^2}$$

The structure of $M(t)$ reflects two effects: first, measurement accuracy degrades as the engagement progresses due to increased scintillation noise; second, the linear measurement model $\theta \simeq y/V(t_f - t)$ is no longer valid close to the target since θ is not small. The term $R_2/(t_f - t)^2$ effectively down-weights measurements in the region where the linearization breaks down.

Performance Index

The objective is to minimize the expected value of a cost functional that penalizes both the terminal miss distance and the cumulative control effort:

$$J = E\left\{\frac{1}{2}y(t_f)^2 + \frac{b}{2}\int_0^{t_f} a_P(t)^2\,dt\right\}$$

This can be expressed in the standard quadratic form:

$$J = E\left[\int_0^{t_f} \left(x_t^T Q_t x_t + u_t^T R u_t\right) dt + x_f^T S_f x_f\right]$$

Matching terms between the two expressions yields the weighting matrices:

$$Q_t = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \qquad R = \frac{b}{2}, \qquad S_f = \begin{bmatrix} \frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

The fact that $Q_t = 0$ indicates that there is no running penalty on the state trajectory. Only the terminal position $y(t_f)$ is penalized through S_f, while control effort is penalized continuously through R.

Initial State and Uncertainty

At the beginning of the engagement, the estimated state is assumed to have zero mean:

$$\hat{x}_0 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

The uncertainty in this initial estimate is captured by the covariance matrix:

$$P_0 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 16 & 0 \\ 0 & 0 & 400 \end{bmatrix}$$

The diagonal structure indicates that the initial errors in y, v, and a_T are assumed to be uncorrelated. The zero variance for y implies that the initial lateral position is known exactly, while significant uncertainty exists in the velocity and target acceleration estimates.

Numerical Data

The simulation employs the following parameter values, corresponding to a typical engagement scenario:

$V = 3000$ ft/s $= 914.4$ m/s	closing velocity
$t_f = 10$ s	engagement duration
$\tau = 2$ s	target maneuver time constant
$W = 100$ m^2/s^5	process noise intensity
$R_1 = 1.5 \times 10^{-5}$ rad^2/s	base measurement noise
$R_2 = 1.67 \times 10^{-3}$ rad^2/s^3	range-dependent noise coefficient
$b = 1.52 \times 10^{-2}$	control weighting factor

LQG Solution Framework

The LQG problem under partial state information is solved via dynamic programming. The solution consists of three components: an optimal feedback law, a state estimator, and an expression for the expected cost.

Optimal Feedback Law

The optimal control is a linear function of the estimated state:

$$u^o(t) = -\underbrace{R^{-1}B^T S_t}_{K_c}\hat{x}_t$$

where $\hat{x}_t$ denotes the state estimate and K_c is the control gain matrix. Note that the admissible control $a_P(t)$ must be a function of the measurement history $\{z_s, 0 \leq s \leq t\}$; in the LQG framework, this requirement is satisfied since $\hat{x}_t$ is produced by the Kalman filter using only past measurements. The matrix S_t satisfies the control Riccati differential equation:

$$-\dot{S}_t = F^T S_t + S_t F - S_t B R^{-1} B^T S_t + Q$$

This equation is integrated backward in time starting from the terminal condition $S(t_f) = S_f$.

State Estimator

The state estimate $\hat{x}_t$ is generated by a Kalman filter:

$$d\hat{x}_t = (F\hat{x}_t + Bu_t^o)\,dt + K_t(dz_t - H_t\hat{x}_t\,dt), \qquad \hat{x}(0) = \hat{x}_0$$

The Kalman gain is given by:

$$K_t = P_t H_t^T M_t^{-1}$$

where P_t is the estimation error covariance, obtained by solving the filtering Riccati equation forward in time:

$$\dot{P}_t = FP_t + P_t F^T - P_t H_t^T M_t^{-1} H_t P_t + GWG^T, \qquad P(0) = P_0$$

Expected Optimal Cost

The expected value of the optimal cost-to-go from time t to t_f is:

$$J^o = \|\hat{x}_t\|^2_{S_t} + \int_t^{t_f} \mathrm{tr}(K_\tau M_\tau K_\tau^T S_\tau)\,d\tau + \int_t^{t_f} \mathrm{tr}(P_\tau Q)\,d\tau + \mathrm{tr}(P_{t_f} S_f)$$

This expression combines the cost associated with the current estimate, the penalty due to estimation uncertainty, and the terminal cost.

Numerical Implementation

The LQG problem was solved by numerically integrating the two Riccati differential equations to obtain $S(t)$ and $P(t)$. From these, the control gain $K_c(t)$ and Kalman gain K_t were derived. The continuous-time formulation was discretized using a time step of $\Delta t = 0.01$ s for numerical integration.

Control Riccati Equation and Feedback Gain

The feedback control law takes the form:

$$u_t^o = -R^{-1}B^T S_t \hat{x}_t = -K_c(t)\hat{x}_t$$

The gain matrix $K_c(t)$ can be written explicitly as:

$$K_c(t) = \frac{2}{b}\begin{bmatrix}0 & 1 & 0\end{bmatrix} S_t = \frac{2}{b}\begin{bmatrix}s_{21}(t) & s_{22}(t) & s_{23}(t)\end{bmatrix}$$

Defining the individual gain components:

$$\begin{bmatrix}k_{c_1}(t) & k_{c_2}(t) & k_{c_3}(t)\end{bmatrix} \triangleq \frac{2}{b}\begin{bmatrix}s_{21}(t) & s_{22}(t) & s_{23}(t)\end{bmatrix}$$

To obtain $S(t)$, we integrate the Riccati equation backward from t_f:

$$-\dot{S}_t = F^T S_t + S_t F - S_t B R^{-1} B^T S_t + Q, \qquad S(t_f) = S_f$$

The backward integration is performed using the following discrete-time update:
for $t = t_f$ **to** t_i**:**

$$\begin{cases} \dot{S}_t = -F^T S_t - S_t F + S_t B R^{-1} B^T S_t - Q \\ S_{t-\Delta t} = S_t - \dot{S}_t \Delta t \end{cases}$$

Figure 2 displays the computed gain components over the engagement interval.

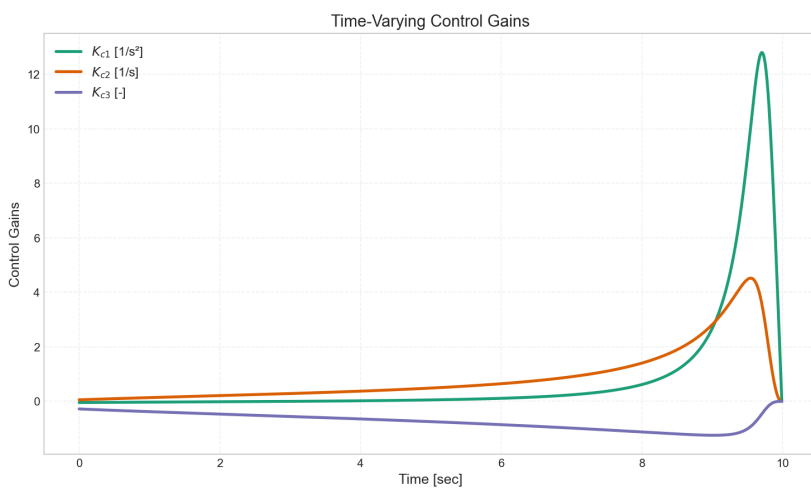

Figure 2: Evolution of control gain components $k_{c_1}(t)$, $k_{c_2}(t)$, $k_{c_3}(t)$ over time.

The gain $k_{c_1}(t)$, associated with position error, grows sharply as $t \to t_f$ before returning to zero at the terminal time. This occurs because the cost function penalizes only the terminal miss distance, causing the controller to act conservatively at first and aggressively near the end. The return to zero at exactly $t = t_f$ is a consequence of the terminal boundary condition in the Riccati equation.

In scalar form, the control law becomes:

$$u_t^o = -(k_{c_1}\hat{y}_t + k_{c_2}\hat{v}_t + k_{c_3}\hat{a}_{T_t})$$

Since this expression requires the estimated state, the Kalman filter must be implemented before the control can be computed.

Estimation Riccati Equation and Kalman Gain

The Kalman gain is derived from the estimation error covariance P_t:

$$K_t = P_t H_t^T M_t^{-1}$$

Expanding this expression with the explicit matrix dimensions:

$$K_t = P_t H_t^T M_t^{-1} = \begin{bmatrix} p_{11}(t) & p_{12}(t) & p_{13}(t) \\ p_{21}(t) & p_{22}(t) & p_{23}(t) \\ p_{31}(t) & p_{32}(t) & p_{33}(t) \end{bmatrix} \begin{bmatrix} \frac{1}{V(t_f - t)} \\ 0 \\ 0 \end{bmatrix} M_t^{-1}$$

This simplifies to:

$$K_t = \begin{bmatrix} p_{11}(t) \\ p_{21}(t) \\ p_{31}(t) \end{bmatrix} \frac{M_t^{-1}}{V(t_f - t)}$$

Substituting the expression for M_t, the Kalman gain components are:

$$K_t = \begin{bmatrix} K_1 \\ K_2 \\ K_3 \end{bmatrix} = \frac{t_f - t}{V(R_1(t_f - t)^2 + R_2)} \begin{bmatrix} p_{11}(t) \\ p_{21}(t) \\ p_{31}(t) \end{bmatrix}$$

The covariance matrix P_t is obtained by integrating the filtering Riccati equation forward from $t = 0$:

$$\dot{P}_t = FP_t + P_t F^T - P_t H_t^T M_t^{-1} H_t P_t + GWG^T, \qquad P(0) = P_0$$

The forward integration scheme is:
for $t = t_i$ **to** t_f**:**

$$\begin{cases} \dot{P}_{t_k} = FP_{t_k} + P_{t_k} F^T - P_{t_k} H_t^T M_t^{-1} H_t P_{t_k} + GWG^T \\ P_{t_{k+1}} = P_{t_k} + \Delta t \cdot \dot{P}_{t_k} \end{cases}$$

Figure 3 shows the time evolution of the covariance elements p_{11}, p_{21}, and p_{31}.

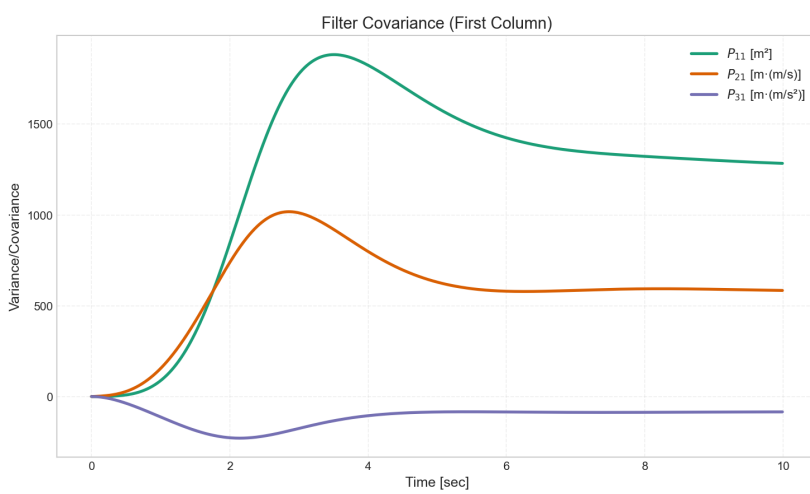

Figure 3: Evolution of covariance matrix elements $p_{11}(t)$, $p_{21}(t)$, $p_{31}(t)$.

The position variance p_{11} starts at zero (known initial position), increases to a peak as process noise accumulates, then decreases as measurements improve the estimate. The covariance p_{21} follows a similar pattern, while p_{31} becomes negative, reflecting the correlation structure between position and target acceleration errors.

The resulting Kalman gain components $K_1(t)$, $K_2(t)$, $K_3(t)$ are shown in Figure 4.

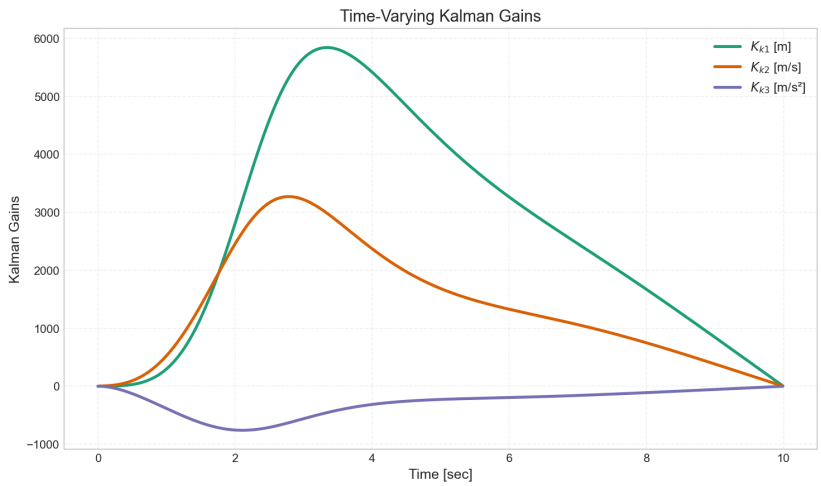

Figure 4: Kalman gain components $K_1(t)$, $K_2(t)$, $K_3(t)$ computed from $P(t)$.

Initially, when uncertainty is high, the Kalman gains generally take larger values in magnitude as the filter relies more on measurements to correct the state estimate. The position gain K_1 reaches a peak in the early phase of the engagement. As more observations are processed, the estimation improves and the gains decrease. Note that K_3 may become negative due to the sign of the cross-covariance term $p_{31}(t)$. Near the final time, however, the measurement noise intensity $M(t) = R_1 + \frac{R_2}{(t_f - t)^2}$ grows unboundedly. Consequently, the filter reduces its reliance on measurements and the gains approach zero as $t \to t_f$. This adaptive behavior balances trust between the dynamic model and the noisy observations throughout the engagement.

Closed-Loop Dynamics

With the Riccati solutions in hand, we now assemble the complete closed-loop system. The optimal control is:

$$u_t^o = -K_c(t)\hat{x}_t$$

The estimated state evolves according to the Kalman filter equation:

$$d\hat{x}_t = (F\hat{x}_t + Bu_t^o)\,dt + K_t(dz_t - H_t\hat{x}_t\,dt)$$

Substituting the observation model $dz_t = H_t x_t\,dt + dm_t$ and rearranging:

$$d\hat{x}_t = [(F - K_tH_t)\hat{x}_t + Bu_t^o]\,dt + K_tH_tx_t\,dt + K_t\,dm_t$$

The true state satisfies the stochastic differential equation:

$$dx_t = (Fx_t + Bu_t^o)\,dt + G\,d\beta_t$$

Substituting the optimal control $u_t^o = -K_c(t)\hat{x}_t$, the coupled system can be written in Itô form as:

$$\begin{cases} d\hat{x}_t = [(F - K_tH_t - BK_c)\hat{x}_t + K_tH_tx_t]\,dt + K_t\,dm_t \\ dx_t = [Fx_t - BK_c\hat{x}_t]\,dt + G\,d\beta_t \end{cases}$$

where $K_c = R^{-1}B^TS_t$ is the control gain.
This system is integrated forward in time to obtain the trajectories of both the actual and estimated states under optimal feedback.

Numerical Results

We now present the simulation outcomes based on the equations developed above. The plots compare the true state trajectories with their estimates over the engagement duration for a single realization of the stochastic process.

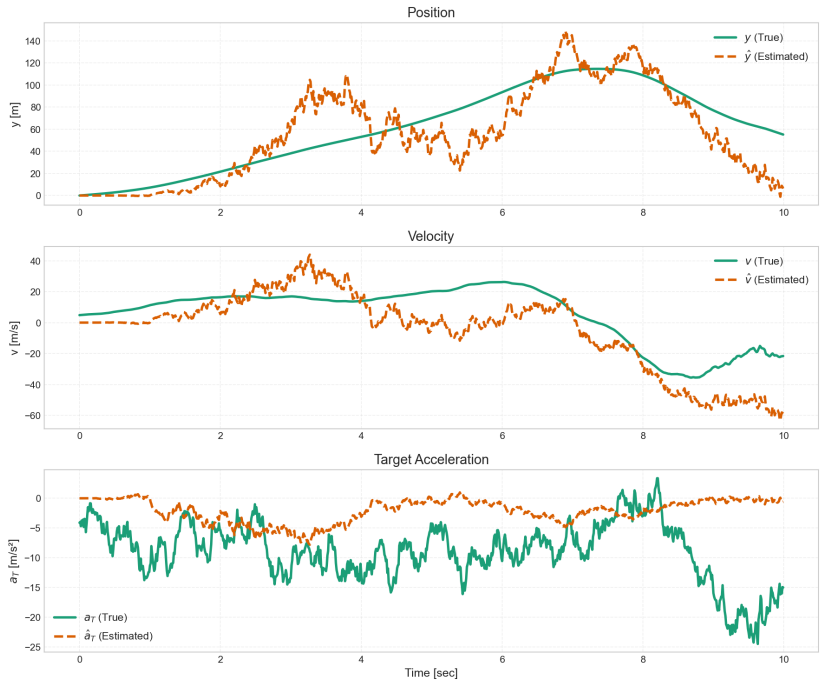

Figure 5: Actual and estimated states over time: lateral displacement y, velocity v, and target acceleration a_T.

The figure shows a single stochastic realization of the engagement. In this particular run, the lateral displacement y increases initially before the controller drives it back toward zero. The final miss distance for this run lies within the expected statistical range (see Monte Carlo analysis below). The Kalman filter estimates for position and velocity track the true states reasonably well despite measurement noise. The target acceleration estimate, however, shows limited tracking capability since a_T is not directly observable and must be inferred from the position measurements.

The corresponding optimal control signal is displayed below.

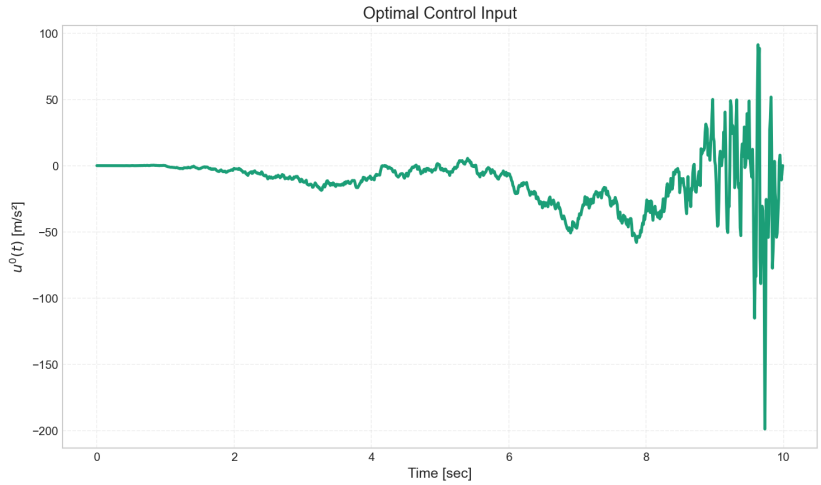

Figure 6: Optimal pursuer acceleration $u^o(t)$ over the engagement.

The control effort remains moderate during the early phase, as the cost function emphasizes terminal accuracy rather than trajectory regulation. Near the final time, the controller acts aggressively with large acceleration commands to

eliminate residual miss distance. This pattern is consistent with the structure of the weighting matrices, where only the final position error is penalized, and with the sharp increase in control gains observed in Figure 2.

Analysis of the Cost-to-Go

The expected optimal cost from any time t to the final time is given by:

$$J^o = \|\hat{x}_t\|^2_{S_t} + \int_t^{t_f} \mathrm{tr}(K_\tau M_\tau K_\tau^T S_\tau)\, d\tau + \int_t^{t_f} \mathrm{tr}(P_\tau Q)\, d\tau + \mathrm{tr}(P_{t_f} S_f)$$

The cost-to-go exhibits a general downward trend over time. However, due to the stochastic nature of the problem, the trajectory shows fluctuations rather than a strictly monotonic decrease. These fluctuations arise from the state-dependent term $\|\hat{x}_t\|^2_{S_t}$, which varies with the realized noise.

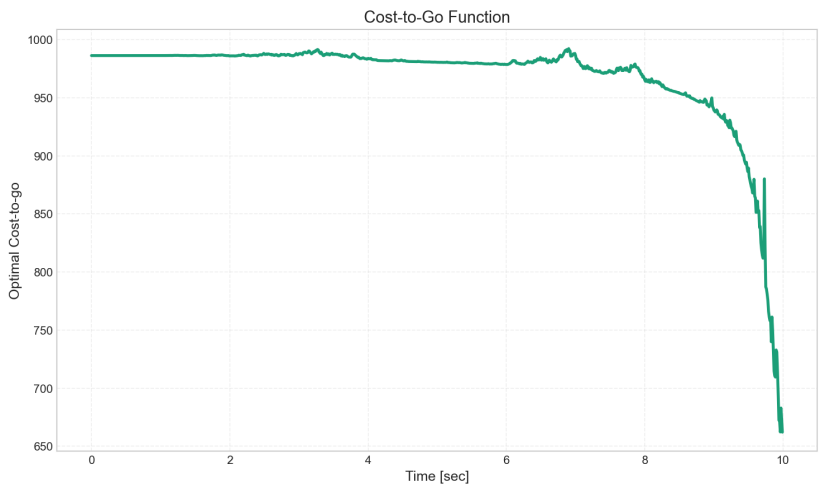

Figure 7: Cost-to-go function over time for a single simulation run.

At $t = 0$, the cost $J^o(0)$ is nearly deterministic since $\|\hat{x}_0\|^2_{S_0} = 0$ due to the initial condition $\hat{x}_0 = [0\ 0\ 0]^T$. As time progresses, the state-dependent component introduces variability while the deterministic components decrease smoothly.

The decrease in the cost-to-go can be understood by examining its individual components. Since $Q = 0$ in this problem, the state is not penalized along the trajectory and the integral $\int_t^{t_f} \mathrm{tr}(P_\tau Q)\, d\tau$ vanishes identically.

The individual components of the cost function are shown below.

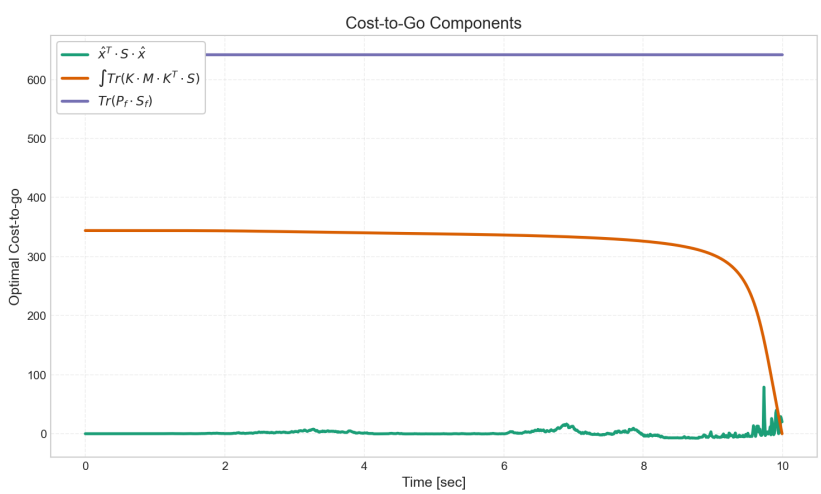

Figure 8: Cost-to-go components over time.

With $Q = 0$, the optimal cost expression reduces to three main parts:

- $\mathrm{tr}(P_{t_f} S_f)$ represents the residual uncertainty cost at the terminal time. This deterministic term accounts for the estimation error at final time and contributes the largest portion to the total cost. It remains constant throughout the simulation since it depends only on the final covariance.
- $\int_t^{t_f} \mathrm{tr}(K_\tau M_\tau K_\tau^T S_\tau)\, d\tau$ decreases smoothly to zero as the integration interval shrinks. This term quantifies the cumulative effect of measurement noise on the cost and can be computed offline since it does not depend on the realized state trajectory.
- $\|\hat{x}_t\|^2_{S_t}$ remains small for most of the engagement but shows spikes near the final time as the control gains increase. This is the only component that depends on the actual state realization.

Monte Carlo Analysis

To evaluate the statistical performance of the guidance system, a Monte Carlo simulation with 500 independent runs was conducted. Each run used different realizations of process noise β and measurement noise m, while the control and Kalman gains remained fixed (computed once from the Riccati equations).

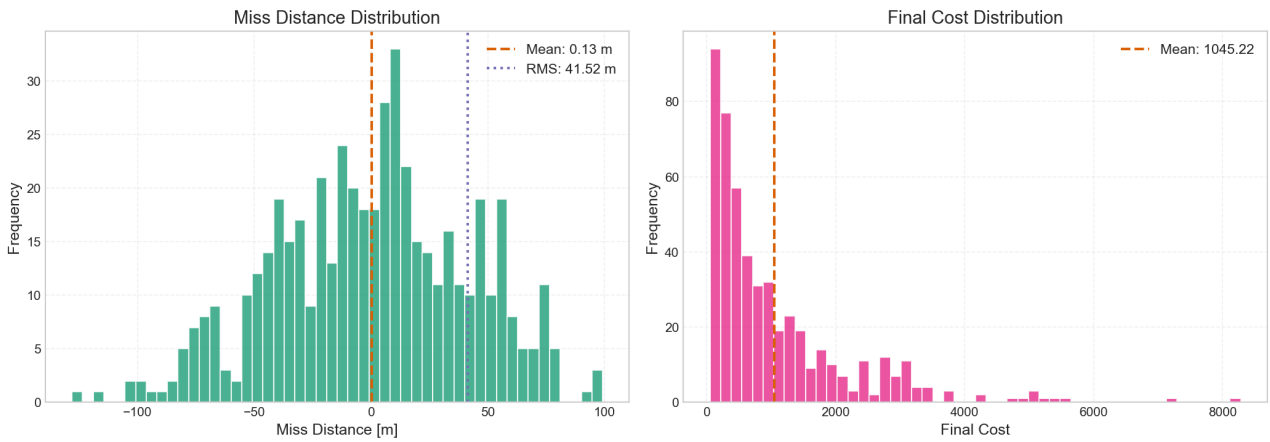

Figure 9: Distribution of miss distance and final cost over 500 Monte Carlo runs.

The results are summarized in the following table:

Metric	Value	Interpretation
Mean miss distance	0.13 m	Near zero (unbiased controller)
RMS miss distance	41.5 m	Standard deviation due to noise
Mean final cost	1045	Consistent with cost-to-go decomposition

The near-zero mean miss distance confirms that the LQG controller is **unbiased**: on average, the pursuer hits the target. The symmetric distribution centered at zero in the left histogram visually confirms this property.

The relatively large RMS value of 41.5 m reflects the inherent uncertainty in the problem rather than any deficiency in the controller. Several factors contribute to this variance:

- High process noise intensity ($W = 100$ m^2/s^5) causes significant random target maneuvers
- Measurement noise increases as the engagement progresses due to the structure of $M(t)$
- The target acceleration a_T is not directly observable, limiting estimation accuracy

Regarding the final cost distribution, the final-cost histogram is right-skewed: most runs achieve relatively low terminal cost, while a small number of rare runs produce much larger costs, resulting in a long tail. This behavior is expected due to the quadratic cost terms (in particular $y(t_f)^2$), which amplify occasional large miss-distance events and shift the mean to the right. Despite these challenges, the LQG controller achieves the optimal performance given the available information. No other linear controller using the same measurements could achieve a lower expected cost.

Discussion and Conclusions

This project developed a comprehensive simulation framework for the Linear Quadratic Gaussian (LQG) control problem in the context of missile guidance. The proposed approach integrates state estimation via Kalman filtering with optimal feedback control, enabling systematic analysis of system performance under both process noise and measurement uncertainty. The numerical results validate the theoretical framework and demonstrate successful controller operation.

The Kalman filter exhibited effective performance in reconstructing the state trajectory despite the presence of stochastic disturbances. The time-varying Kalman gain K_t, computed from the estimation error covariance P_t, effectively balanced reliance on the dynamic model and incoming measurements. When measurement noise was low, the filter weighted observations heavily; as the noise intensity increased near the final time due to the structure of $M(t)$, the filter shifted trust toward the model predictions. This adaptive mechanism is clearly visible in Figure 4, where all Kalman gains approach zero as $t \to t_f$.

On the control side, the optimal feedback law $u_t^o = -K_c(t)\hat{x}_t$ successfully minimized the cost function. The control gain $K_c(t)$, obtained by solving the Riccati equation backward in time from the terminal condition $S(t_f) = S_f$, dynamically adjusted the control input based on the estimated state. During the early phase of the engagement, the controller applied moderate corrections since the cost function does not penalize trajectory deviations. As the system approached the final time t_f, the gains increased sharply (Figure 2) to eliminate residual miss distance.

The Monte Carlo analysis provided the most important validation of the controller's effectiveness. Over 500 independent runs, the mean miss distance was near zero, confirming that the controller is unbiased. The RMS miss distance, while substantial, represents the irreducible uncertainty inherent in the stochastic problem formulation. This variance is consistent with the terminal uncertainty term $\mathrm{tr}(P_{t_f} S_f)$, which accounts for the estimation error at intercept.

The evolution of the cost-to-go function (Figures 7 and 8) provided additional insight into the controller's behavior. While the total cost showed fluctuations due to stochastic effects, the deterministic components decreased smoothly over time. The dominant contribution to the final cost came from the terminal uncertainty term $\mathrm{tr}(P_{t_f} S_f)$, highlighting that estimation accuracy at the final time is the primary limiting factor in this guidance problem.

In conclusion, the LQG framework provides a powerful and systematic methodology for optimal control under uncertainty. The separation principle allows independent design of the estimator and controller, while guaranteeing overall optimality when combined. The results of this project validate the theoretical foundations of LQG control and demonstrate its practical applicability to guidance and control problems in aerospace engineering. The Monte Carlo analysis confirms that the implemented controller achieves optimal performance: it is unbiased and minimizes the expected quadratic cost given the available noisy measurements.